

Generative AI Tools

B. Tech I Year I Semester					Common to AI, AIML			
Code	Category	Hours / Week			Credits	Marks		
EMA1109	Exploratory (PC)	L	T	P	C	CIE	SEE	Total
		1	0	4	3	80	20	100

Course Outcomes

At the end of this course, students will be able to:

1. Identify real world problems and select appropriate Gen AI tools.
2. Apply prompt engineering techniques to generate accurate and context aware output.
3. Identify and Compare various Gen AI tools.
4. Create text, images, presentations and other digital content.
5. Employ Gen AI Tools ethically, responsibly and securely.

List of Experiments

Week 1

- a) Generating a Story: Use a generative AI text tool to create a short story and a poem from a prompt, written in the style of a specified author.
- b) Reimagining Book Covers: Generate book covers for classic literature using AI, emphasizing thematic representation.

Week 2

- a) Study Guides Creation: Generate comprehensive study guides based on course materials.
- b) Design Digital Artwork: Create digital art from textual descriptions using AI.

Week 3

- a) Conceptualize Science Fiction Worlds: Generate detailed descriptions of science fiction worlds using AI and visualize the same.
- b) Creating Fashion Designs: Use AI to generate fashion designs focusing on historical accuracy of traditional garments from various cultures.

Week 4

- a) Summarizing Articles: Use AI to summarize a lengthy article, focusing on key points and main ideas.
- b) Create AI-Generated Movie Scenes: Use AI to generate images depicting scenes from described movie scripts, focusing on capturing the emotion and action of the narrative.

Week 5

- a) Translating Languages: Use AI to translate a paragraph from one language to another.
- b) Brand Development Using AI: Create a complete brand identity package including logo, color scheme, and brand guidelines.

Week 6

- a) Generate Legal Documents: Use AI to draft basic legal documents like non-disclosure agreements or lease agreements.
- b) Animation Character Design: Use AI to design characters for an animation character, focusing on unique traits and expressions.

Week 7

- a) Generating Cooking Recipes: Create unique cooking recipes based on specified ingredients using AI.
- b) AI-Generated Character Design: create visual aids for subjects like history, science, and literature, bringing textbook content to life.

Week 8

- a) Creating a Travel Itinerary: Design a detailed travel itinerary for a 3-day trip using AI.
- b) Custom Jewelry Designs: Use AI to design personalized jewelry based on individual preferences and trends.

Week 9

- a) Personal Budgeting: Generate a personalized budget plan based on income and spending habits using AI.
- b) Architectural Rendering: Generate 3D architectural renderings of buildings or interior spaces based on provided specifications and styles using AI.

Week 10

- a) Composing Music: Compose a musical composition, complete with lyrics, piano, and guitar notes, utilizing AI software tailored to a specified genre or mood.
- b) Fashion Tech Integration: Create designs that integrate technology, such as wearable tech, using AI.

Week 11

- a) Generate a Novel Outline: Create a detailed script for short film including characters, plot, and setting using AI.
- b) AI-Generated Landscape Design: Utilize AI to design landscapes for virtual environments or architectural projects.

Week 12

- a) Generating Workout Routines: Develop personalized workout routines based on individual fitness goals, preferences, and available equipment using AI.
- b) Digital Costume Design: Use AI to design detailed costumes for films, theater, or video games.

Week 13

- a) Crafting Gardening Plans: Generate gardening plans for a backyard or indoor garden, considering factors like climate, available space, and preferred plants.
- b) Space Exploration Visualization: Generate visualizations of distant planets, galaxies, or cosmic phenomena based on scientific data and descriptions using AI.

Week 14

- a) Develop Presentation Slides: Create presentation slides on a given topic using AI
- b) AI-Powered Archaeological Reconstructions: Generate visual and narrative reconstructions of archaeological sites or historical events using AI

Reference Books

1. Stephen Lucci, Danny Kopec, Artificial Intelligence in the 21st Century - A Living Introduction, Mercury Learning and information, 2nd Edition, 2016.